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COLLEGE OF ENGINEERING
Autonomous
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Kacharam, Shamshabad, Hyderabad- 501 218, Telangana, India.

Department of Computer Science and Engineering

IV B. TECH I SEMESTER CSE-D

BATCH 2023-27	PROJECT WORK PHASE-I 2026-2027	ACADEMIC YEAR 2026-2027

Batch Id **:** 23MPCS-D13

Title of the Project **:** Predictive Adaptive Load Balancing for Distributed Systems Using Online Performance Modeling

TEAM MEMBERS

S. No.	Roll Number	Student Name	Mail Id	Contact Number
1	23881A05M8	KUCHI HARSHA VARDHAN	harshakuchi02@gmail.com	9381742258
2	23881A05K3	AKEPOGU ALLEN CYRUS PRAKASH	allen.akepogu2005@gmail.com	8106194100
3	24885A0527	CHATRE RITIKA	ritikachatre@gmail.com	7780770129

GUIDE DETAILS

Name of the Guide	:	Donagani Ramakrishna
Designation	:	Assistant Professor
Department	:	Computer Science and Engineering
Mail Id	:	ramakrishna1909@vardhaman.org
Contact Number	:	+91 9390230903
Area of interest	:	Post Quantum Cryptography

Signature of the Projects In charge

Signature of the Guide with Date

Signature of the Project Coordinator

Batch Id **:** 23MPCS-D13

Title of the Project **:** Predictive Adaptive Load Balancing for Distributed Systems Using Online Performance Modeling

Objectives

1. To design and develop an adaptive load balancer that dynamically distributes requests across multiple servers based on predicted performance.

2. To implement an online performance modeling system that continuously monitors server metrics and predicts future workload conditions.

3. To evaluate the proposed predictive load balancing approach against conventional algorithms using latency, throughput, and resource utilization metrics.

Outcomes

1. A functional predictive load balancing system capable of improving request distribution in a distributed environment.

2. Reduced response time and improved resource utilization compared to traditional load balancing techniques under varying workloads.

3. A validated performance analysis demonstrating the effectiveness of online performance modeling for adaptive decision-making in distributed systems.

ABSTRACT

Distributed systems rely on efficient load balancing to ensure high availability, low response time, and optimal utilization of computing resources. Traditional load balancing algorithms, such as Round Robin and Least Connections, make routing decisions based only on the current state of the servers, which may lead to performance degradation under dynamic workloads. This project proposes a **Predictive Adaptive Load Balancing for Distributed Systems Using Online Performance Modeling** approach that predicts future server performance and dynamically distributes incoming client requests to the most suitable backend server.

The system consists of multiple backend servers, a custom load balancer, a monitoring module, and a machine learning-based prediction model. The monitoring module continuously collects runtime metrics such as CPU utilization, memory usage, active

connections, request rate, queue length, and response time from each server. These metrics are processed by an online performance model that predicts future server latency and workload. Based on these predictions, the load balancer adaptively routes incoming requests to the server expected to provide the best performance, thereby reducing response time, preventing server overload, and improving overall system throughput.

The project is implemented using **C++** for the load balancer, backend servers, and networking components, utilizing **POSIX sockets**, **multithreading**, and **TCP/HTTP communication**. The performance prediction model is developed using **Python** with **Scikit-learn** (Random Forest or XGBoost) and integrated with the C++ system for real-time inference. Performance evaluation is carried out by comparing the proposed approach with conventional load balancing algorithms using metrics such as average response time, throughput, latency, and server resource utilization under different workload conditions.

The proposed system aims to demonstrate that predictive, adaptive load balancing using online performance modeling can significantly enhance the efficiency, scalability, and reliability of distributed systems compared to traditional static or reactive load balancing techniques.

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Date of Submission:

Signature of the Guide with Date	***Signature of the HOD, CSE***